

07.1

A student investigated the activity of the enzyme polyphenol oxidase (PPO) in apple tissue.

When apple tissue is exposed to air, PPO catalyses a reaction between colourless phenol compounds in apple tissue and oxygen.

Figure 7 shows the reaction.

Figure 7



The student measured the time taken for the brown pigment to appear in two apple varieties (**D** and **E**).

Method

1. Cut a 1 cm cube of apple tissue from variety **D**.
2. Put the cube on a plate and leave the plate at 30 °C
3. Measure the time for the brown pigment to appear.
4. Repeat steps 1 to 3 two more times.
5. Repeat steps 1 to 4 with apple tissue from variety **E**.

The student obtained the results shown below, but did not record the data in a results table.

Variety **D** = 15 min 50 s, 18 min, 14 min 30 s

Variety **E** = 6 min 30 s, 8 min, 7 min

In the box labelled **Figure 8**, design a suitable results table.

- Enter the student's results into the table.
- Calculate the mean results and include these in the table.
- Use 1 decimal place for both mean results.

[3 marks]

Figure 8

0 7 . 2 Suggest and explain why the results for variety **D** are different from the results for variety **E**.

[2 marks]

Question 7 continues on the next page

Turn over ►

0 7 . 3

The student repeated the investigation but made **one** change to the method used to prepare the apple tissue.

The student then observed shorter times for the brown pigment to appear in both apple varieties.

Suggest the change the student made to the method of preparing the apple tissue.

Explain why the brown pigment appeared in a shorter time.

Do **not** suggest using a different volume or mass of apple.

[3 marks]

Change to method _____

Explanation _____

0 7 . 4

The student wants to change the procedure to obtain a measure of PPO activity either in terms of how much substrate is used or how much product is produced.

Which change in procedure will provide a successful measure of PPO activity for the student?

Tick (✓) **one** box.

[1 mark]

Measure the increase in oxygen concentration in the air around the cube of apple tissue on a plate.

☐

Measure the intensity of brown colour by comparing apple tissue with a colour chart showing a range of apple tissues of known pigment concentration.

☐

Measure the intensity of colour produced from brown apple tissue in a biuret test.

☐

Measure the percentage of light transmitted through a cube of brown apple tissue using a colorimeter.

☐☐

